Psychology is the scientific study of the mind and behavior.[1][2] Its subject matter includes the behavior of humans and nonhumans, both conscious and unconscious phenomena, and mental processes such as thoughts, feelings, and motives. Psychology is an academic discipline of broad scope, crossing the boundaries between the natural and social sciences. Biological psychologists seek an understanding of the emergent properties of brains, linking the discipline to neuroscience. As social scientists, psychologists aim to understand the behavior of individuals and groups.[3][4]

A professional practitioner or researcher involved in the discipline is called a psychologist. Some psychologists can also be classified as behavioral or cognitive scientists. Some psychologists attempt to understand the role of mental functions in individual and social behavior. Others explore the physiological and neurobiological processes that underlie cognitive functions and behaviors.

As part of an interdisciplinary field, psychologists are involved in research on perception, cognition, attention, emotion, intelligence, subjective experiences, motivation, brain functioning, and personality. Psychologists' interests extend to interpersonal relationships, psychological resilience, family resilience, and other areas within social psychology. They also consider the unconscious mind.[5] Research psychologists employ empirical methods to infer causal and correlational relationships between psychosocial variables. Some, but not all, clinical and counseling psychologists rely on symbolic interpretation.

While psychological knowledge is often applied to the assessment and treatment of mental health problems, it is also directed towards understanding and solving problems in several spheres of human activity. By many accounts, psychology ultimately aims to benefit society.[6][7][8] Many psychologists are involved in some kind of therapeutic role, practicing psychotherapy in clinical, counseling, or school settings. Other psychologists conduct scientific research on a wide range of topics related to mental processes and behavior. Typically the latter group of psychologists work in academic settings (e.g., universities, medical schools, or hospitals). Another group of psychologists is employed in industrial and organizational settings.[9] Yet others are involved in work on human development, aging, sports, health, forensic science, education, and the media.

Etymology and definitions

The word *psychology* derives from the Greek word *psyche*, for spirit or soul. The latter part of the word *psychology* derives from -λογία *-logia*, which means "study" or "research".[10] The word psychology was first used in the Renaissance.[11] In its Latin form *psychiologia*, it was first employed by the Croatian humanist and Latinist Marko Marulić in his book *Psichiologia de ratione*

animae humanae (*Psychology, on the Nature of the Human Soul*) in the decade 1510–1520[11][12] The earliest known reference to the word *psychology* in English was by Steven Blankaart in 1694 in *The Physical Dictionary*. The dictionary refers to "Anatomy, which treats the Body, and Psychology, which treats of the Soul."[13]

Ψ (*psi*), the first letter of the Greek word *psyche* from which the term psychology is derived, is commonly associated with the field of psychology.

In 1890, William James defined *psychology* as "the science of mental life, both of its phenomena and their conditions."[14] This definition enjoyed widespread currency for decades. However, this meaning was contested, notably by John B. Watson, who in 1913 asserted the methodological behaviorist view of psychology as a purely objective experimental branch of natural science, the theoretical goal of which "is the prediction and control of behavior."[15] Since James defined "psychology", the term more strongly implicates scientific experimentation.[16][15] Folk psychology is the understanding of the mental states and behaviors of people held by ordinary people, as contrasted with psychology professionals' understanding.[17]

History

Main article: History of psychology

For a chronological guide, see Timeline of psychology.

The ancient civilizations of Egypt, Greece, China, India, and Persia all engaged in the philosophical study of psychology. In Ancient Egypt the Ebers Papyrus mentioned depression and thought disorders.[18] Historians note that Greek philosophers, including Thales, Plato, and Aristotle (especially in his *De Anima* treatise),[19] addressed the workings of the mind.[20] As early as the 4th century BCE, the Greek physician Hippocrates theorized that mental disorders had physical rather than supernatural causes.[21] In 387 BCE, Plato suggested that the brain is where mental processes take place, and in 335 BC Aristotle suggested that it was the heart.[22]

In China, the foundations of psychological thought emerged from the philosophical works of ancient thinkers like Laozi and Confucius, as well as the teachings of Buddhism.[23] This body of knowledge drew insights from introspection, observation, and techniques for focused thinking and behavior. It viewed the universe as comprising physical and mental realms, along with the interplay between the two.[24] Chinese philosophy also emphasized purifying the mind in order to increase virtue and power. An ancient text known as *The Yellow Emperor's Classic of Internal Medicine* identifies the brain as the nexus of wisdom and sensation, includes theories of personality based on yin–yang balance, and analyzes mental disorder in terms of physiological and social disequilibria. Chinese scholarship that focused on the brain advanced during the Qing dynasty with the work of Western-educated Fang Yizhi (1611–1671), Liu Zhi (1660–1730),

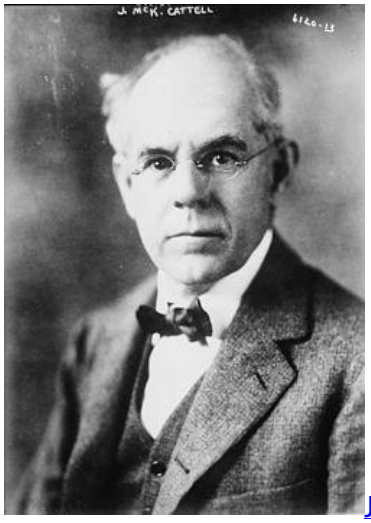
and Wang Qingren (1768–1831). Wang Qingren emphasized the importance of the brain as the center of the nervous system, linked mental disorder with brain diseases, investigated the causes of dreams and insomnia, and advanced a theory of hemispheric lateralization in brain function.[25]

Influenced by Hinduism, Indian philosophy explored distinctions in types of awareness. A central idea of the *Upanishads* and other Vedic texts that formed the foundations of Hinduism was the distinction between a person's transient mundane self and their eternal, unchanging soul. Divergent Hindu doctrines and Buddhism have challenged this hierarchy of selves, but have all emphasized the importance of reaching higher awareness. Yoga encompasses a range of techniques used in pursuit of this goal. Theosophy, a religion established by Russian-American philosopher Helena Blavatsky, drew inspiration from these doctrines during her time in British India.[26][27]

Psychology was of interest to Enlightenment thinkers in Europe. In Germany, Gottfried Wilhelm Leibniz (1646–1716) applied his principles of calculus to the mind, arguing that mental activity took place on an indivisible continuum. He suggested that the difference between conscious and unconscious awareness is only a matter of degree. Christian Wolff identified psychology as its own science, writing *Psychologia Empirica* in 1732 and *Psychologia Rationalis* in 1734. Immanuel Kant advanced the idea of anthropology as a discipline, with psychology an important subdivision. Kant, however, explicitly rejected the idea of an experimental psychology, writing that "the empirical doctrine of the soul can also never approach chemistry even as a systematic art of analysis or experimental doctrine, for in it the manifold of inner observation can be separated only by mere division in thought, and cannot then be held separate and recombined at will (but still less does another thinking subject suffer himself to be experimented upon to suit our purpose), and even observation by itself already changes and displaces the state of the observed object."

In 1783, Ferdinand Ueberwasser (1752–1812) designated himself *Professor of Empirical Psychology and Logic* and gave lectures on scientific psychology, though these developments were soon overshadowed by the Napoleonic Wars.[28] At the end of the Napoleonic era, Prussian authorities discontinued the Old University of Münster.[28] Having consulted philosophers Hegel and Herbart, however, in 1825 the Prussian state established psychology as a mandatory discipline in its rapidly expanding and highly influential educational system. However, this discipline did not yet embrace experimentation.[29] In England, early psychology involved phrenology and the response to social problems including alcoholism, violence, and the country's crowded "lunatic" asylums.[30]

Beginning of experimental psychology



James McKeen Cattell, the first psychologist in the United States

Wilhelm Wundt (seated), a German psychologist, with colleagues in his psychological laboratory, the first of its kind, c. 1880

One of the dogs used in Russian psychologist Ivan Pavlov's experiment with a surgically implanted cannula to measure saliva, preserved in the Pavlov Museum in Ryazan, Russia

Philosopher John Stuart Mill believed that the human mind was open to scientific investigation, even if the science is in some ways inexact.[31] Mill proposed a "mental chemistry" in which elementary thoughts could combine into ideas of greater complexity.[31] Gustav Fechner began conducting psychophysics research in Leipzig in the 1830s. He articulated the principle that human perception of a stimulus varies logarithmically according to its intensity.[32]:61 The principle became known as the Weber–Fechner law. Fechner's 1860 *Elements of Psychophysics* challenged Kant's negative view with regard to conducting quantitative research on the mind.[33][29] Fechner's achievement was to show that "mental processes could not only be given numerical magnitudes, but also that these could be measured by experimental

methods."[29] In Heidelberg, Hermann von Helmholtz conducted parallel research on sensory perception, and trained physiologist Wilhelm Wundt. Wundt, in turn, came to Leipzig University, where he established the psychological laboratory that brought experimental psychology to the world. Wundt focused on breaking down mental processes into the most basic components, motivated in part by an analogy to recent advances in chemistry, and its successful investigation of the elements and structure of materials.[34] Paul Flechsig and Emil Kraepelin soon created another influential laboratory at Leipzig, a psychology-related lab, that focused more on experimental psychiatry.[29]

James McKeen Cattell, a professor of psychology at the University of Pennsylvania and Columbia University and the co-founder of *Psychological Review*, was the first professor of psychology in the United States.[35]

The German psychologist Hermann Ebbinghaus, a researcher at the University of Berlin, was a 19th-century contributor to the field. He pioneered the experimental study of memory and developed quantitative models of learning and forgetting.[36] In the early 20th century, Wolfgang Kohler, Max Wertheimer, and Kurt Koffka co-founded the school of Gestalt psychology of Fritz Perls. The approach of Gestalt psychology is based upon the idea that individuals experience things as unified wholes. Rather than reducing thoughts and behavior into smaller component elements, as in structuralism, the Gestaltists maintain that whole of experience is important, "and is something else than the sum of its parts, because summing is a meaningless procedure, whereas the whole-part relationship is meaningful."[37]

Psychologists in Germany, Denmark, Austria, England, and the United States soon followed Wundt in setting up laboratories.[38] G. Stanley Hall, an American who studied with Wundt, founded a psychology lab that became internationally influential. The lab was located at Johns Hopkins University. Hall, in turn, trained Yujiro Motora, who brought experimental psychology, emphasizing psychophysics, to the Imperial University of Tokyo.[39] Wundt's assistant, Hugo Münsterberg, taught psychology at Harvard to students such as Narendra Nath Sen Gupta—who, in 1905, founded a psychology department and laboratory at the University of Calcutta.[26] Wundt's students Walter Dill Scott, Lightner Witmer, and James McKeen Cattell worked on developing tests of mental ability. Cattell, who also studied with eugenicist Francis Galton, went on to found the Psychological Corporation. Witmer focused on the mental testing of children; Scott, on employee selection.[32]:60

Another student of Wundt, the Englishman Edward Titchener, created the psychology program at Cornell University and advanced "structuralist" psychology. The idea behind structuralism was to analyze and classify different aspects of the mind, primarily through the method of introspection.[40] William James, John Dewey, and Harvey Carr advanced the idea of functionalism, an expansive approach to psychology that

underlined the Darwinian idea of a behavior's usefulness to the individual. In 1890, James wrote an influential book, *The Principles of Psychology*, which expanded on the structuralism. He memorably described "stream of consciousness." James's ideas interested many American students in the emerging discipline.[40][14][32]: 178–82 Dewey integrated psychology with societal concerns, most notably by promoting progressive education, inculcating moral values in children, and assimilating immigrants.[32]: 196–200

A different strain of experimentalism, with a greater connection to physiology, emerged in South America, under the leadership of Horacio G. Piñero at the University of Buenos Aires.[41] In Russia, too, researchers placed greater emphasis on the biological basis for psychology, beginning with Ivan Sechenov's 1873 essay, "Who Is to Develop Psychology and How?" Sechenov advanced the idea of brain reflexes and aggressively promoted a deterministic view of human behavior.[42] The Russian-Soviet physiologist Ivan Pavlov discovered in dogs a learning process that was later termed "classical conditioning" and applied the process to human beings.[43]